

UMPHANG, Thailand

WMO: 483850

Lat: 16.0253N

Lon: 98.8600E

Elev: 1509

StdP: 13.91

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	50.1	53.0	44.8	46.3	68.0	48.0	52.4	66.2	8.3	82.8	7.2	82.7	0.0	N/A	0.226

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	24.3	96.0	71.1	94.0	70.9	92.1	71.3	77.3	85.7	76.6	84.8	76.3	84.2	1.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	140.3	80.3	74.6	137.2	79.7	74.2	135.3	79.3	42.0	86.1	41.4	85.1	41.0	84.8	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.9	4.8	2.1	DB	45.3	99.1	3.8	1.8	42.6	100.4	40.4	101.4	38.3	102.4	35.5	103.7
			WB	44.4	80.8	3.7	2.3	41.7	82.5	39.6	83.8	37.5	85.1	34.8	86.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.8	71.3	74.3	78.0	81.1	80.6	79.1	77.6	77.4	78.1	77.8	74.9	71.4
	DBStd	4.07	3.66	3.24	2.79	2.30	2.38	2.05	1.76	1.69	1.73	1.86	3.17	4.00
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	10	4	1	0	0	0	0	0	0	0	0	1	4
	CDD50	9779	659	680	868	933	949	873	855	849	843	860	746	664
	CDD65	4314	198	261	403	483	484	423	390	384	393	395	297	203
	CDH74	33550	1939	2840	4238	4735	4012	2922	2178	1993	2282	2561	2180	1670
	CDH80	12638	707	1358	2247	2462	1616	818	420	350	547	831	766	516
(14) Wind	WSAvg	0.3	0.3	0.4	0.5	0.4	0.2	0.2	0.1	0.1	0.1	0.2	0.3	0.3
(15) Precipitation	PrecAvg	58.8	0.4	0.4	2.1	3.5	7.6	7.7	9.9	10.0	10.2	5.8	1.1	0.4
	PrecMax	75.9	1.3	2.6	5.4	11.8	14.7	11.7	20.5	15.6	16.9	12.9	5.3	3.0
	PrecMin	37.8	0.0	0.0	0.0	0.5	2.7	3.8	5.6	5.5	4.0	0.3	0.0	0.0
	PrecStd	9.4	0.5	0.6	1.8	2.5	3.1	2.2	3.4	2.5	3.1	2.8	1.2	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.3	94.5	97.2	99.0	97.0	91.5	88.2	87.6	89.2	90.0	90.0	89.2
		MCWB	68.7	68.2	68.7	72.1	74.5	75.7	75.5	76.1	76.1	74.8	74.0	71.7
	2%	DB	88.4	92.5	95.2	96.8	93.2	89.0	85.9	85.6	87.0	88.2	88.3	87.1
		MCWB	68.9	68.4	68.8	72.1	75.1	75.7	75.2	75.7	75.9	74.8	73.3	71.1
	5%	DB	86.5	90.8	93.6	94.5	90.5	86.7	84.1	83.6	85.2	86.7	86.9	85.4
		MCWB	68.5	68.7	68.8	72.7	75.3	75.5	74.9	75.1	75.5	74.5	72.6	70.5
	10%	DB	84.6	88.7	91.6	92.0	88.1	84.6	82.4	81.9	83.4	85.1	85.0	83.2
		MCWB	67.9	68.4	68.9	73.1	75.3	75.3	74.6	74.7	75.2	74.3	72.1	69.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.9	74.5	74.8	77.4	77.9	77.7	76.9	77.3	77.6	77.4	76.7	75.5
		MCDB	83.8	87.0	87.0	88.8	87.2	85.6	83.6	84.0	85.1	85.8	85.8	84.4
	2%	WB	72.2	72.7	73.6	76.1	77.0	76.8	76.2	76.3	76.7	76.5	75.5	73.6
		MCDB	82.2	85.5	86.4	87.9	86.8	84.3	82.5	82.6	83.9	84.6	84.6	82.9
	5%	WB	71.0	71.5	72.7	75.3	76.3	76.2	75.6	75.6	76.1	75.8	74.5	72.1
		MCDB	81.0	83.9	84.9	87.3	85.7	83.4	81.5	81.5	82.9	83.5	83.3	81.3
	10%	WB	69.8	70.3	71.8	74.5	75.7	75.6	75.1	75.1	75.5	75.1	73.3	70.9
		MCDB	79.7	82.7	83.2	86.1	84.6	82.3	80.7	80.6	81.8	82.3	81.5	80.0
(35) Mean Daily Temperature Range	5% DB	MDBR	29.1	31.5	29.1	24.3	16.5	12.4	10.7	10.5	12.2	16.3	21.6	25.6
		MCDBR	31.2	33.7	32.5	28.0	20.6	15.7	13.3	13.2	14.8	18.9	23.4	26.2
		MCWBR	13.2	12.1	9.4	7.4	5.7	4.7	4.5	4.8	5.4	7.4	10.1	11.6
	5% WB	MCDBR	24.6	27.9	25.3	22.9	17.0	13.5	11.5	11.7	13.4	16.0	20.3	22.8
		MCWBR	11.0	11.3	8.3	7.3	5.6	4.8	4.6	4.8	5.3	6.8	9.8	10.9
(40) Clear-Sky Solar Irradiance	taub		0.448	0.534	0.657	0.585	0.469	0.422	0.416	0.419	0.437	0.510	0.450	0.443
	taud		2.160	1.945	1.675	1.902	2.279	2.434	2.449	2.429	2.362	2.103	2.245	2.201
	Ebn at Noon		266	247	222	239	265	275	277	279	275	251	263	263
	Edh at Noon		46	60	80	64	43	37	36	37	40	50	42	43
(44) All-Sky Solar Radiation	RadAvg		1506	1671	1746	1865	1609	1376	1263	1257	1391	1477	1527	1464
	RadStd		68	70	111	113	149	137	129	105	95	86	86	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.23	+0.36	N/A	N/A	N/A

Nomenclature: See separate page